KATULA VAISHNAVI

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SUMMARY

B.Tech Data Science student (2027) with hands-on experience in Artificial Intelligence, Machine Learning, Data Science, Data Analysis, Predictive Modeling, and Data Engineering. Experienced in data gathering, data exploration, data transformation, ETL pipelines, model evaluation, and analytics-ready dataset preparation using Python, SQL, Scikit-learn, TensorFlow, and cloud technologies. Passionate about solving business problems through data-driven decision-making, statistical analysis, and scalable AI solutions.

EDUCATION

Malla Reddy Deemed to be University Hyderabad, Telangana
B.Tech in Data Science (CGPA: 7.81) *Aug 2023 – Jun 2027*

EXPERIENCE

Edunet Foundation Hyderabad
AI and Cloud Intern *Jun 2025 – Aug 2025*
- Built multi-model Disease Prediction system benchmarking Logistic Regression, Random Forest & XGBoost.
- Performed feature engineering, handled missing/imbalanced data & hyperparameter tuning for better recall.
- Evaluated models with precision, recall, F1, ROC-AUC and presented SHAP-based clinical explainability.
- Deployed the model via Flask API and documented the pipeline using AWS cloud fundamentals.

TECHNICAL SKILLS

Languages: Python, SQL
ML & AI: Scikit-learn, PyTorch, TensorFlow, RAG, Prompt Engineering, Generative AI (LLMs)
Data, DB & Tools: Pandas, NumPy, ChromaDB, LangChain, NoSQL, ETL, Power BI, Matplotlib, Seaborn
Engineering: FastAPI, Flask, Cloud (Basics), Streamlit, React, Docker, Git, AWS

PROJECTS

AmazonQA-RAG | Python, QLoRA, FAISS, LangChain, RAGAS GitHub
- Fine-tuned LLM via QLoRA (4-bit NF4 + LoRA) for low-resource instruction adaptation on Amazon Q&A data.
- Built FAISS-backed RAG pipeline with sentence-transformers; benchmarked Base/RAG/Finetuned on RAGAS.
- Optimized retrieval latency & context handling, achieving 28% better answer relevancy over base RAG.
- Added streaming response support and citation highlighting in the Streamlit interface for better UX.

Retail Sales Data Warehouse | Python, PostgreSQL, Airflow, dbt, Docker, Power BI GitHub
- Built star-schema DWH (SCD Type 2) processing 10K rows / $2.3M revenue with 10 data quality checks.
- Delivered 3 business marts (RFM, inventory turnover, daily KPIs), cutting reporting time from 4 hrs to 8 min.
- Automated incremental loads & dbt tests with Dockerized Airflow, reducing manual work by 90%.
- Created interactive Power BI dashboards with row-level security for different business stakeholders.

Matrix Factorisation Recommender | Python, PyTorch, SVD, NCF, BPR GitHub
- Implemented & benchmarked 5 models (SVD, NCF, BPR vs Popularity, Item-kNN) on NDCG@10 / MAP@10.
- Designed cold-start evaluation and ran ablations on embedding size, negative sampling & dropout via FastAPI.
- Achieved 15% NDCG@10 lift over strongest baseline and packaged inference service with Docker.
- Added real-time recommendation endpoint with caching and A/B testing support for model comparison.

CERTIFICATIONS

- Google GenAI AI/ML – Hack2Skill, 2025
- Introduction to Technology and AWS – Coursera, 2025
- Microsoft Azure AI-900 – Certiport / PearsonVUE, 2023
- Infosys Pragati Cohort – SpringBoard, 2025
- AWS Learning: Generative AI – AWS Skill Builder, 2026